

# Introduction to Phonology

## Week 4

### Distinctive Features

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# Last week

- Phonotactics,
- alternations

# This week

- Phonological universals,
- distinctive features

# Phonological universals

- Do languages vary without limit (Edward Sapir)?
- Are there patterns in languages that are recurrent?

# Phonological universals

- Phonologies of different languages resemble each other.
  - Universal Grammar
    1. Some claim that UG is programmed in our genes,
    2. others claim that the biology of the brain, mouth and ear are responsible for the similarities among languages,
    3. yet others claim that languages are very similar because some changes in language are more likely than others.

# Formalism

- [bɛæk] ~ [bɛɛgə],
- [hʊnt] ~ [hʊndə],
- [lɒp] ~ [lɒbəs],
- [haʊs] ~ [hɔɪzə].

Is this one process or four?

# Formalism

The final consonant in [bɛɛk], [hʊnt], [lop] and [haus] is pronounced without vocal fold vibration, the corresponding consonant in the inflected forms is pronounced with vocal fold vibration.

# Formalism

- One process? We only need one statement that captures all cases at the same time (No word in German ends in a voiced obstruent).
- Four processes? We describe the same process four times (No German word ends in [g]; No German word ends in [d]; No German word ends in [b]; No German word ends in [z]).



# Distinctive features

All segments are defined by their acoustic and articulatory properties. These properties are their **features**. Since these features are the smallest phonological constituents that can distinguish words they are called **distinctive features**.

# Distinctive features

## Hypotheses concerning distinctive features

- Contrasts, classes of sounds and alternations can be described with a **universal, closed set of phonetic parameters**:  
Distinctive features.
- Features are binary. Their value can be + or –.

# Features

## Manner features

The feature [syllabic] divides the sounds in consonants and vowels.

**artic** [+syllabic] sounds are the core of a syllable

**acous** [+syllabic] sounds have higher amplitudes than neighboring sounds.

Vowels are [+syllabic] and all other sounds—from glides to stops—are [-syllabic].

# Features

## Manner features: [+syllabic]

The German word *Familie* can be pronounced in two ways: [famɪːljə] and [famɪːliə]. The difference between the [j] and the [i] can be understood as a difference in the value of [syll] for the segment [i]. If it is [+syll] it is pronounced as [i], if it is [-syll] it is pronounced as [j].

I said ‘can be understood’ because it is better to analyze this variation as a consequence of syllable structure.

# Features

## Manner features

The feature [consonantal] differentiates sounds on the basis of their usual position in a syllable.

**artic** [-cons] sounds have no vocal tract constriction

**acous** [-cons] sounds have high amplitude resonance

Segments that are [+cons] appear usually in the onset of a syllable: liquids, nasals and obstruents. Other segments—glides and vowels—are [-cons].

# Features

## Manner features

Sonorants allow an unimpeded flow of air so that spontaneous voicing is possible (Odden, 2005).

**artic** [-sonorant] sounds have a build-up of pressure behind an oral constriction

**acous** [-sonorant] sounds have lower amplitude resonance

Segments that are [-son] impede the flow of air either completely (stops) or to such an extent that turbulences arise (fricatives).

# Features

## Manner features

Sounds that are [-approximant] have a closure in the mouth.

**artic** [-approximant] sounds have a complete closure

**acous** [-approximant] sounds have an even lower amplitude resonance

Nasals, fricatives and stops are [-approximant].

# Features

## Manner features

Natural classes of manner features are often contiguous on the Sonority Hierarchy.

| <b>Vowels</b>  | <b>Glides</b> | <b>Liquids</b> | <b>Nasals</b>  | <b>Obstruents</b> |
|----------------|---------------|----------------|----------------|-------------------|
| [+syllabic]    | [-syllabic]   |                |                |                   |
| [-consonantal] |               | [+consonantal] |                |                   |
| [+approximant] |               |                | [-approximant] |                   |
| [+sonorant]    |               |                |                | [-sonorant]       |



# Features

## Manner features

The feature [nasal] distinguishes between nasal sounds and non-nasal sounds.

**artic** [+nasal] sounds have an open velum

**acous** [+nasal] sounds have nasal resonance

# Features

## A practical example

Many languages have nasalized vowels: English and Zeeuwsch-Vlaams have them allophonically (English [mæ̃n] ‘man’, [θĩŋ] ‘thing’ and [bãm] ‘bomb’ (Hayes, 2009); Zeeuwsch-Vlaams [mẽns] ‘human being’, [vlõms] ‘flemish’, E.A. van de Vijver, p.c.). The feature [nasal] allows us to refer to the class of all nasal sounds. We don’t have to specify that the same process applies to labial nasals, coronal nasals and dorsal nasals. This is good.

# Features

## Manner features

All sounds that allow air to escape along the sides of the tongue are laterals.

**artic** [+lateral] sounds have a vocal tract closure along a center line, open at the side

**acous** [+lateral] sounds have a particular resonance

The speech sounds [l, r] are laterals.

# Features

## Manner features

German has affricates—sounds that are relatively rare; they start out as stop and end as a fricative. These sounds have the feature [delayed release].

**artic** [+delayed release] sounds have a stop phase that slowly gives way to a fricative phase

**acous** [+delayed release] period of friction upon release

The affricates [tʃ] and [pʃ] are affricates. Why are these considered one sound and not two?

# Features

## Manner features

Speech that are [+cont] have a constriction that does not block the air flow (Odden, 2005).

**artic** [+continuant] sounds have an uninterrupted air flow

**acous** [+continuant] sounds have no silence

Fricatives are [+cont].

# Features

## Manner features

We can now distinguish between stop, fricatives and affricates.

| <b>stops</b>       | <b>affricates</b>  | <b>fricatives</b> |
|--------------------|--------------------|-------------------|
| [-continuant]      |                    | [+continuant]     |
| [-delayed release] | [+delayed release] |                   |

# Features

## Manner features

Now we can define segments with features:

vowel = [+syllabic]

glide = [-syllabic, -consonantal]

liquid = [+consonantal, +approximant, +lateral]

nasal = [-approximant, +sonorant, +nasal]

obstruent = [-sonorant]

# Feature

## Manner features

In Mandarin Chinese only these segments may be aspirated: [p], [t], [k], [tʰ], [tʂʰ], [tʃʰ]. Describe the phonemes using their manner of articulation and use features to characterize their natural class.



# Features

## Laryngeal features

Laryngeal features describe the state of the glottis.

**artic** [+voice] sounds have vibrating vocal folds

**acous** [+voice] sounds are periodic

Are vowels and sonorants [+voice]? (Hint: [+voice] is a  
**distinctive**

# Features

## Laryngeal features

Speech sounds that are released into a glottal fricative are aspirated.

**artic** [+spread glottis] sounds have a large glottal opening

**acous** [+spread glottis] extended period of voicelessness upon release

The first obstruents in the German pair [pəkən] and [bəkən] contrast in [spread glottis], rather than in [voice]. The Dutch pair [pakən] and [bakən] differ in [voice], rather than in [spread glottis].

# Features

## Laryngeal features

Speech sounds that have very tight vocal folds, which causes an irregular crackling sound are creaky.

**artic** [+constricted glottis] sounds have constricted vocal folds

**acous** [+constricted glottis] irregular glottal pulses

This is what it sounds like.

# Features

## Major place features

- [labial] sounds have the lips as active articulator.
- [coronal] sounds have the tip of the tongue as active articulator.
- [dorsal] sounds have the tongue body as active articulator.
- [pharyngeal] sounds have the pharynx as active articulator.
- [laryngeal] sounds are articulated at the glottis.

# Features

## Subsidiary place features

Coronal sounds come in several flavors.

- [+anterior] sounds are produced with the tongue front at or in front of the alveolar ridge.
- [-anterior] sounds are made with the tongue front behind the alveolar ridge.

# Features

## Subsidiary place features

Coronal sounds come in several flavors.

- [+anterior] sounds are produced with the tongue front at or in front of the alveolar ridge.
- [-anterior] sounds are made with the tongue front behind the alveolar ridge.
- [+distributed] sounds are made with a long constriction.
- [-distributed] sounds are made with a short constriction.

# Features

## Subsidiary place features

Coronal sounds come in several flavors.

- [+anterior] sounds are produced with the tongue front at or in front of the alveolar ridge.
- [-anterior] sounds are made with the tongue front behind the alveolar ridge.
- [+distributed] sounds are made with a long constriction.
- [-distributed] sounds are made with a short constriction.
- [+strident] sounds have high amplitude friction.
- [-strident] sounds have low amplitude friction.

# Features

## Subsidiary place features

Both [+anterior] and [-anterior] can be made with the **tip** of the tongue or the **blade** of the tongue.

|                  |                   |                                     |
|------------------|-------------------|-------------------------------------|
| [+distributed]   | =                 | blade (laminal)                     |
| [-distributed]   | =                 | tip (apical)                        |
| lamino-dentals   | [t̪, d̪, θ, ð, ɲ] | [+coronal, +anterior, +distributed] |
| apico-alveolars  | [t, d, s, z, n]   | [+coronal, +anterior, -distributed] |
| palato-alveolars | [tʃ, dʒ, ʃ, ʒ]    | [+coronal, -anterior, +distributed] |
| retroflexes      | [ɭ, ɻ, ʂ, ʐ, ɻ]   | [+coronal, -anterior, -distributed] |

[+coronal] fricatives can be [+strident]: [s, z, t̪s̪, d̪z̪, ʃ, ʒ, tʃ, dʒ].  
English plural allomorphy!



# Features

## Subsidiary place features

- [round] = articulated by rounding the lips
- [labiodental] = articulated by bringing the lower lip to the upper teeth.

# Features

## Subsidiary place features

|                    |                                                                                                  |                                 |
|--------------------|--------------------------------------------------------------------------------------------------|---------------------------------|
| plain labials      | [p, b, m, $\phi$ , $\beta$ ]                                                                     | [+labial, -round, -labiodental] |
| plain labiodents   | [f, v]                                                                                           | [+labial, -round, +labiodental] |
| rounded bilabials  | [p <sup>w</sup> , b <sup>w</sup> , m <sup>w</sup> , $\phi$ <sup>w</sup> , $\beta$ <sup>w</sup> ] | [+labial, +round, -labiodental] |
| rounded labiodents | [f <sup>w</sup> , v <sup>w</sup> ]                                                               | [+labial, +round, +labiodental] |
| rounded velars     | [w, k <sup>w</sup> , g <sup>w</sup> , x <sup>w</sup> , $\gamma$ <sup>w</sup> ]                   | [+lab, +dorsal, +ro, -labdent]  |
| labio-vearls       | [ $\widehat{kp}$ , $\widehat{gb}$ ]                                                              | [+lab, +dorsal, -ro, -labdent]  |
| rounded coronals   | [t <sup>w</sup> , d <sup>w</sup> , s <sup>w</sup> , z <sup>w</sup> , r <sup>w</sup> ]            | [+lab, +cor, +ro, -labdent]     |

# Features

## Vowel features

We have just seen this feature for consonants.

**artic** [+round] sounds have pursed lips

**acous** [+round] sounds have lowered resonances

An ongoing research question is whether this feature [round] is the same one for vowels and consonants.

# Features

## Vowel features

**artic** [+high] sounds have a raised tongue body

**acous** [+high] sounds have a low first formant

High vowels in German are [i, y, u].

# Features

## Vowel features

**artic** [+low] sounds have a lowered tongue body

**acous** [+low] sounds have high first formant

Low vowels in German are [a, ɔ]. (Some say there is only one low vowel [a], which can be short or long.)

# Features

## Vowel features

**artic** [+back] sounds have a tongue body that is back from its neutral position

**acous** [+back] sounds have F1 and F2 close together

Back vowels in German are [u, ʊ, o, ɔ, ɑ].

# Features

## Vowel features

**artic** [+advanced tongue root (ATR)] sounds have a tongue root which is pulled forward

**acous** [+advanced tongue root (ATR)] sounds have a lowered F1 in comparison to their [-ATR] counterparts.

The vowels in German that are [+ATR] are [i, y, e, ø, a, o, u] and the vowels that are [-ATR] are [ɪ, ʏ, ɛ, œ, ɔ, ʊ].

# Assignment

- Question 5 page 272 Zsiga (2013).
- Question 6 page 272 Zsiga (2013).



# References

Hayes, B. (2009). *Introductory Phonology*. Wiley-Blackwell.

Odden, D. (2005). *Introducing phonology*. Cambridge university press.

Zsiga, E. C. (2013). *The Sounds of Language: An Introduction to Phonetics and Phonology*. Wiley-Blackwell.